E211 CA1 SAMPLE REPORT (1-Stage Healthcare System)

Background

A community health clinic runs a Health Screening Counter where patients come for basic checks (e.g. blood pressure, height/weight, simple questions about lifestyle). This counter is operated throughout the morning session.

Patients arrive individually, take a queue number, and wait to be called by the nurse at the screening counter. After the screening is completed, patients leave the system (For this assignment, I ignore what happens after that).

This report is to model and document this Health Screening Counter as a single-stage queuing system.

Part 1: System Description – One Stage – Health Screening Counter

a. Build a FlexSim simulation model to simulate the queuing system you described in Part 1.
   i. In this healthcare queuing system, the customers are the patients who come to the clinic for basic health screening. Each patient arrives, takes a queue number, and waits for their turn to be served.
   ii. The server is the nurse at the Health Screening Counter. The nurse's job is to carry out simple checks such as measuring blood pressure, height and weight, and asking a few basic questions about the patient's health. For this model, we assume there is only one nurse providing this service.
   iii. The queue is the line of patients waiting to be served by the nurse. When the nurse is busy with another patient, new arrivals will join the queue and wait.

The queue follows a First-Come, First-Served (FCFS) rule, meaning the patient who has been waiting the longest will be served next.

iv. The main performance concern in this system is the waiting time of patients and the workload of the nurse. If many patients arrive within a short period, the queue can grow long, causing long waiting times and possibly frustration for patients. At the same time, if the nurse is busy almost all the time (high utilisation), it may lead to fatigue and reduced quality of service. Therefore, my simulation study will measure the following:

a) How long patients wait on average?

b) How many patients are usually in the queue? and

c) How heavily the nurse is utilised during the session?

b) **Screenshot of ChatGPT prompt and response**.

Prompt: Generate a simple scenario for a 1-stage Healthcare queuing system. There should be one nurse serving at the counter. Include 20 sets of randomly generated data for the inter-arrival times and service times.

Sample Data (20 Patients)

All times in **minutes**.
Each row = 1 patient.

Patient	Inter-arrival Time	Service Time (Nurse)
1	1.6	18.7
2	0.7	0.2
3	4.2	9.8
4	0.3	1.7
5	3.1	0.8
6	1.8	0.6
7	0.2	1.8

Patient	Inter-arrival Time	Service Time (Nurse)
8	2.8	8.5
9	0.2	1.0
10	2.3	4.4
11	0.3	5.1
12	0.4	2.3
13	2.2	4.0
14	7.0	0.3
15	0.5	0.3
16	1.0	1.2
17	3.9	5.7
18	11.8	2.8
19	3.4	1.9
20	2.0	4.4

Part 2: FLEXSIM MODEL

Apply FLEXSIM, the clinic model is built as follows:

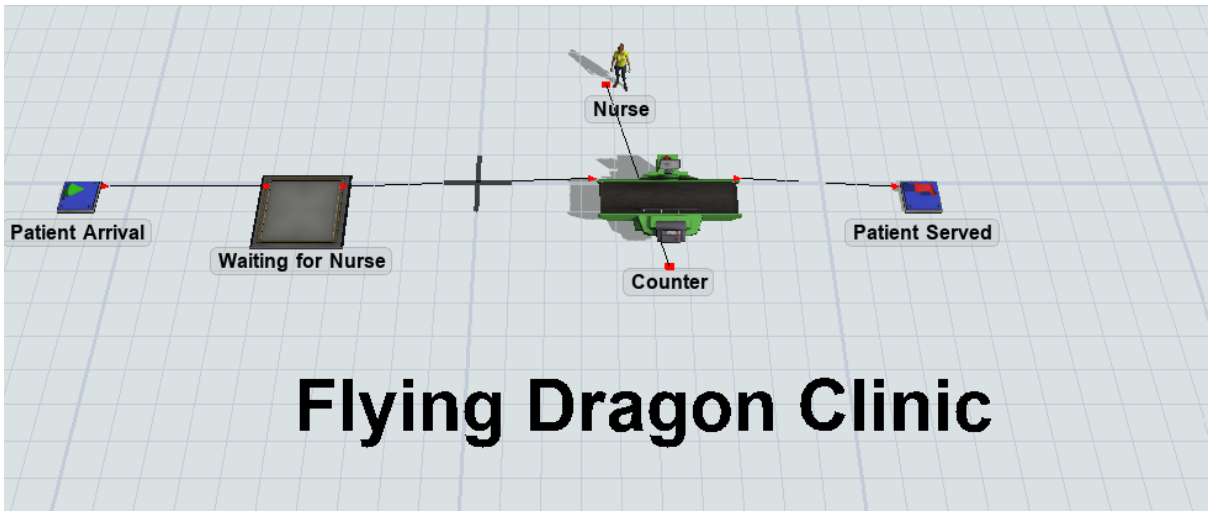


Figure 1: Flexsim Model

a) Source (Patient Arrival) → Queue (Waiting for Service) → Processor (Nurse) → Sink (Patient Served).

Apply Fitted Distribution function in FLEXSIM, the inter-arrival time pattern and service time pattern are determined as: exponential (-0.1,2.5) and weilbull(0.09,3.44,0.88) respectively. See Figure 2 and Figure 3.

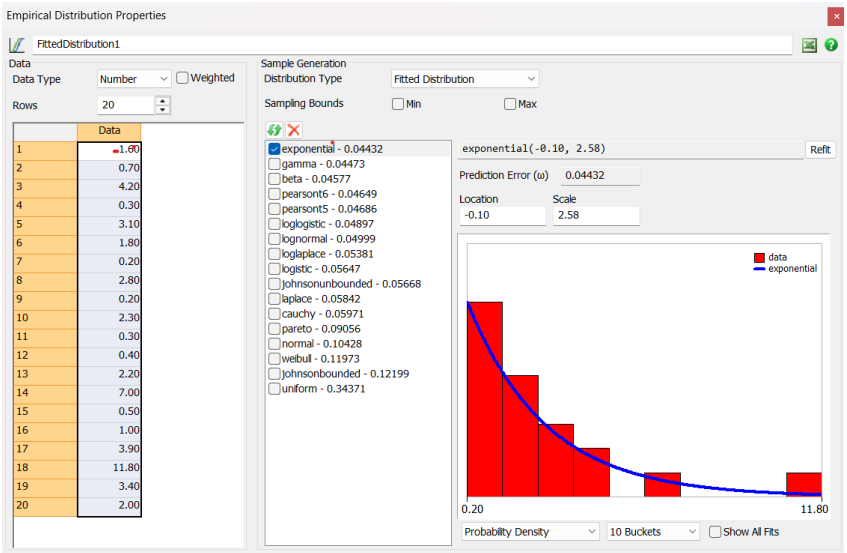


Figure 2: Inter-arrival Fitted Distribution

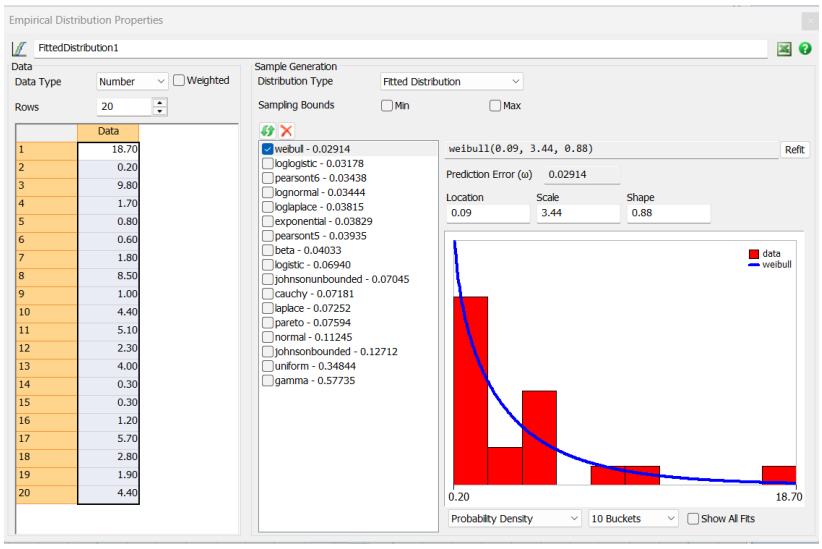


Figure 3: Fitted Distribution Service Time

Study period:

- In this simulation, I assume a half-day (e.g. 4-hour) session for the simulation.
- Set the arrival and service time distributions using your 20 data points.
- Run the simulation for a suitable time (for 4 hours) with 20 replications.
- Collect and report (Dashboard):
    - Average waiting time in queue,
    - Average number of patients in queue,
    - Utilisation of the nurse,
    - Average total time in system.

Waiting time & queue length:
The average waiting time in queue is about 40–41 minutes, with some patients waiting up to ~78 minutes.

At any point in time there are on average about 14 patients waiting, and the queue can go up to 25 patients (and in the Experimenter runs, the average queue length stays extremely high as well).

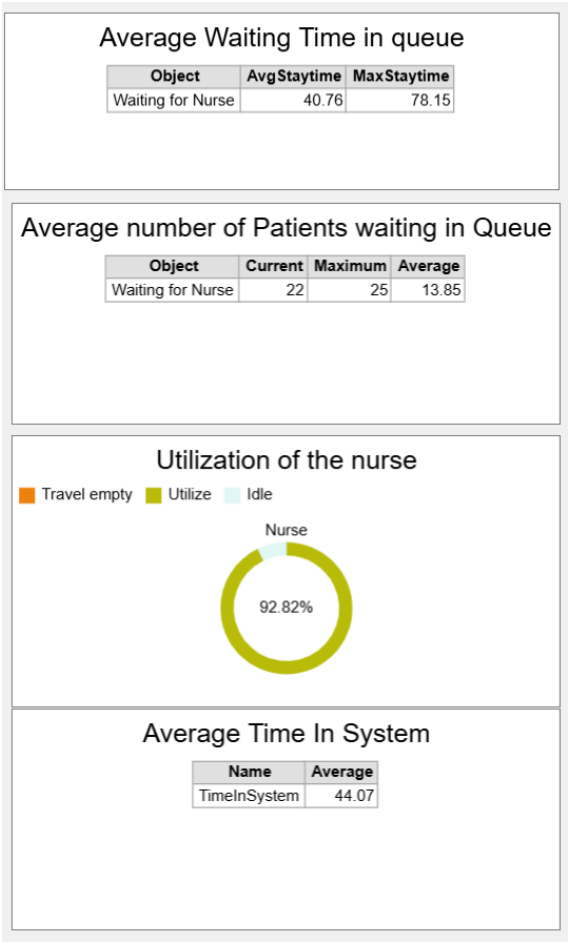


Figure 4: Dashboard for Performance Measures

Average waiting time in queue

	Mean (95% Confidence Interval)	Sample Std Dev	Min	Max
Current Scenario	622.3 ± 21.0	44.9	555.6	757.3

Average number of patients in queue

	Mean (95% Confidence Interval)	Sample Std Dev	Min	Max
Current Scenario	503.1 ± 23.1	49.4	423.0	623.0

Utilisation of Nurse

	Mean (95% Confidence Interval)	Sample Std Dev	Min	Max
Current Scenario	0.997781 ± 0.000744	0.001589	0.994554	0.999956

Average total time in system

	Mean (95% Confidence Interval)	Sample Std Dev	Min	Max
Current Scenario	625.4 ± 21.0	44.9	558.8	760.6

Figure 5: Experimenter Results (20 replications)

6. Comment on the system performance:

From the performance measures, the clinic is clearly overloaded.

- Waiting time & queue length: The average waiting time in queue is about 40–41 minutes, with some patients waiting up to ~78 minutes.
- At any point in time there are on average about 14 patients waiting, and the queue can go up to 25 patients (and in the Experimenter runs, the average queue length stays extremely high as well).
- This indicates that patients spend much more time waiting than being served, which would likely cause dissatisfaction and complaints.

- Nurse utilization - The nurse's utilisation is about 92–93% in the dashboard, and around 99.8% on average in the Experimenter results.
- This means the nurse is almost constantly busy with almost no idle time. In practice this is not sustainable – there is no buffer for breaks, unexpected tasks, or variability in arrivals and service times. Any small increase in demand will immediately create even longer queues.
- Time in system - The average total time in system is about 44 minutes per patient in the dashboard, and more than 10 minutes (if measured in seconds) in the Experimenter output.
- Since the service itself only takes a few minutes, most of this time is spent waiting, not receiving care.

The current clinic configuration with only one nurse at the counter is not performing well. The system is operating very close to full capacity, leading to long queues, long waiting times, and a very high workload for the nurse. From an operations point of view, the clinic would need to either increase capacity (e.g. add another nurse / shorten service time) or reduce demand (e.g. appointment system, limit walk-ins) to achieve acceptable service levels.

Part 3: Propose one simple what-if change: In this assignment, I proposed to add a second nurse during the operations and compare Before vs after (waiting time, queue length, utilisation).

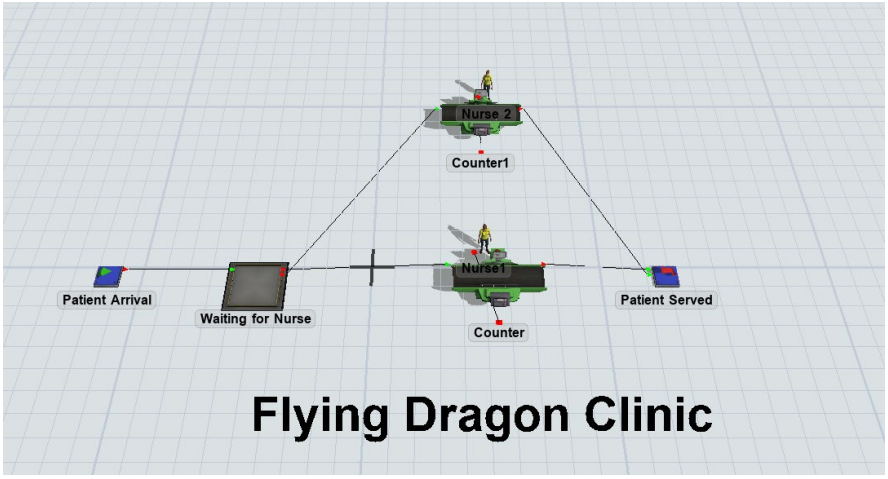
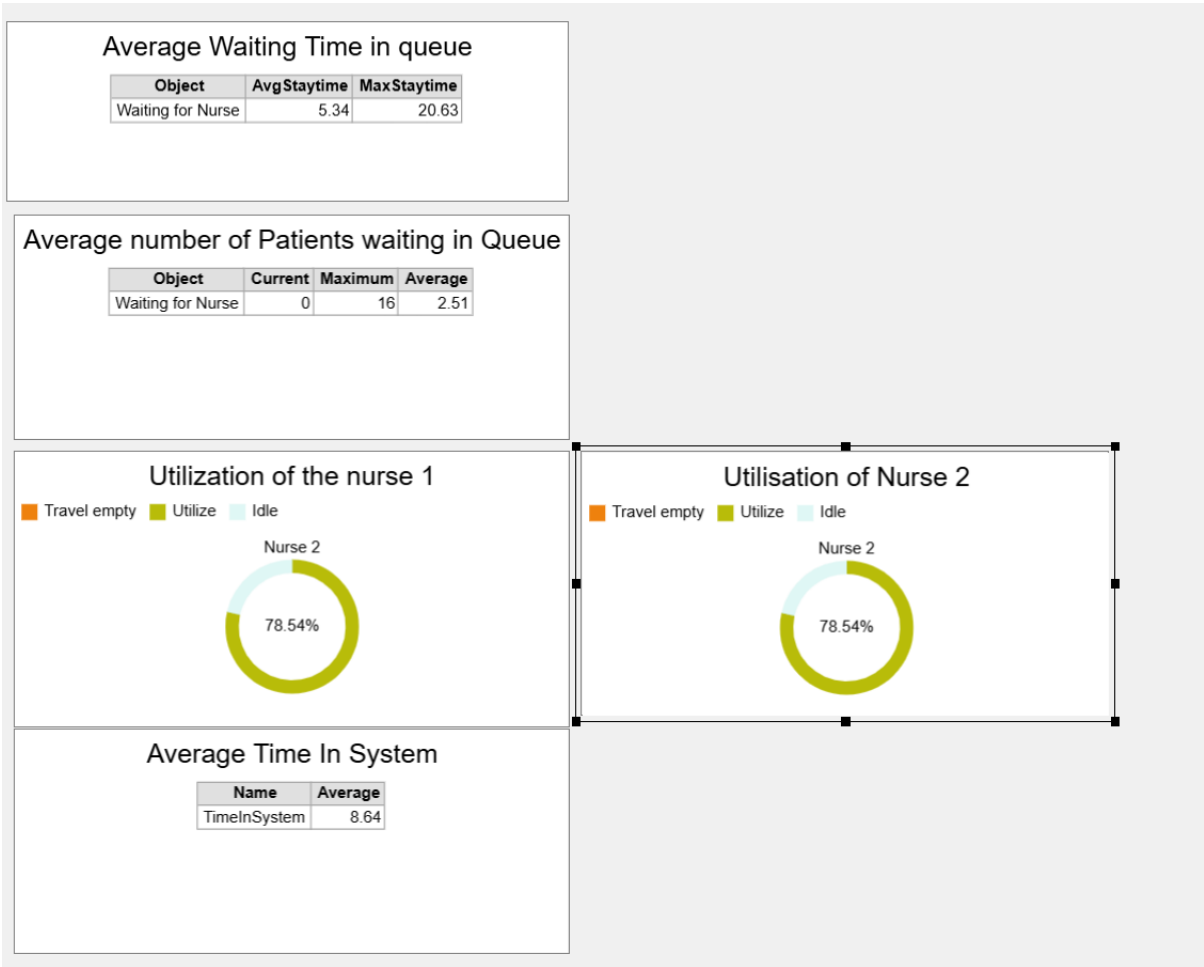


Figure 6: Improved Clinic Model



<u>Figure 7: Dashboard of Performance Measure (Improved model)</u>

After adding **one more nurse** at the screening counter, the clinic performance improved significantly.

- **Waiting time in queue**
  - Before: patients waited on average about **40.8 minutes**.
  - After: average waiting time dropped to about **5.3 minutes** (max about 20.6 minutes).

    → Patients now wait **much shorter**, so the service feels far more acceptable.
- **Average number of patients in queue**
  - Before: average queue length was about **13.9 patients**.
  - After: it reduced to about **2.5 patients** (with current queue often 0).

    → The line is now **much shorter and more stable**, with less crowding.
- **Utilisation of nurses**
  - Before: the single nurse was working at about **92–93% utilisation**, almost fully loaded.
  - After: each nurse is used at about **78.5% utilisation**.

    → Workload is now **shared more evenly** and each nurse has some buffer time, which is more realistic and sustainable.
- **Average time in system**
  - Before: patients spent about **44.1 minutes** in the system.
  - After: this dropped to about **8.6 minutes**.

Overall, adding a second nurse **greatly reduces waiting time and queue length**, while keeping both nurses **busy but not overloaded**. The improved model provides a much better service level and a more manageable working pace for staff.